Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 1st Year 2nd Term Regular Examination, 2023

Math 1217

(Mathematics - II)

Full Marks: 210 Time: 3.0 hrs

N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define limit of a function at a point. Does $f(x) = \frac{x^2-4}{x-2}$ exist at $x = 2$? If exists, find $\lim\limits_{x\to 2} f(x)$. (10)

(b) Discuss the continuity and differentiability of the function $f(x)$ at $x = 0$ (15)
Where $f(x) = \begin{cases} x\sin\left(\frac{1}{x}\right); x \neq 0 \\ 0; \quad x = 0 \end{cases}$

(c) Find the differential coefficient of y where $y = x^{\cos^{-1}x} + (\sin x)^{\ln x}$. (10)

2. (a) Write the statement of Leibnitz's theorem. If $y = \sin^{-1}x$, find the relations among y_{n+2}, y_{n+1}, y_n. (12)

(b) Find the maxima or minima of $\sin x\,(1 + \cos x)$. If exists, find the point of inflexion. (12)

(c) If $y = e^{ax}\sin bx$, find y_n. (11)

3. Integrate the followings:

(a) $\int \frac{dx}{(2x^2+5)\sqrt{x^2-4}}$ (12)

(b) $\int \frac{3\sin x-4\cos x-5}{\cos x-2\sin x+2}dx$ (12)

(c) $\int \frac{dx}{x^4+1}$ (11)

4. (a) Evaluate $\int_0^{\frac{\pi}{2}} \log\cos x\, dx$. (11)

(b) Evaluate $\int_0^{\pi} \frac{x}{a^2\sin^2 x + b^2\cos^2 x}dx$ (12)

(c) Find the area of the segment cutoff from the parabola $y^2 = 4x$ by the straight line $y = 2x$. (12)

Section – B

5 (a) Define O.D.E. and P.D.E. with examples. From a differential equation from the relation $y = Ae^{-x} + Be^{2x} + x^2$ and indicate the order and degree of the obtained differential equation. (12)

(b) Solve $\frac{dy}{dx} = \frac{6x-2y-7}{3x-y+4}$. (11)

(c) Find a particular solution of $y'' - 2y' - 3y = 5$ where $y(0) = 2$ and $y'(0) = 4$. (12)

6. (a) Solve $(2x^2 + y)dx + (x^2y - x)dy = 0$. (13)
   (b) Solve $y^2dx + (3xy - 1)dy = 0$. (11)
   (c) Solve the initial value problem $2x(1 + y)dx + ydy = 0$ where $y(0) = -2$. (11)

7. (a) Solve $y^2dx + (3xy + y^2 - 1)dy = 0$. (11)
   (b) Solve $Cosy\frac{dy}{dx} + \frac{1}{x}siny = 1$. (12)
   (c) Solve $\frac{dy}{dx} = \sin(x + y) + \cos(x + y)$. (12)

8. (a) Solve $(D^2 - 3D + 2)y = sin3x; D = \frac{d}{dx}$. (11)
   (b) Solve $(D^2 + 4D + 4)y = x^3e^{-2x}$. (12)
   (c) Solve $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 2x^2 + cos2x$. (12)

Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 1st Year 2nd Term Examination, 2022

Hum 1217

(Principles of Economics)

Full Marks: 210 Time: 3.0 hrs

N.B. i) Answer any **three** questions from each section in separate script.

ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Elucidate on the basic economic problems with relevant examples. (10)

   (b) Illustrate how the basic economic problems can be solved in different economic systems. (10)

   (c) Use a production possibilities frontier (PPF) to illustrate economy's trade-off between high income and clean environment. (15)

2. Suppose, the market demand for a commodity x is given Q_{dx} =1,20,000-20,000 P_x and market supply equation is given by Q_{sx} =20,000 P_x. (35)

   i. Obtain equilibrium price and quantity,

   ii. If the per unit tax t= $1 is imposed on the goods, what effect does this have on the equilibrium price and quantity?

   iii. What things have been kept constant in this given demand function?

   iv. How do technology and government policy affect the supply of goods and services?

3. (a) Total cost is a combination of fixed cost and variable cost. Explain. (05)

   (b) Explain the importance of capital and role of an entrepreneur in the production process. (10)

   (c) "A rationale producer will always produce in stage-2". Explain with necessary figure. (20)

4. (a) Distinguish between perfect competition market and monopoly market. (05)

   (b) Show with the aid of graph the short-run equilibrium of a firm in perfect competition market. (20)

   (c) At what point do you think a firm should shut down? (10)

Section – B

5. (a) Distinguish between (15)
   i) Normal wage and real wage
   ii) Induced investment and Autonomous investment
   iii) GNP and GDP.

   (b) What are the determinants of investment? Explain. (20)

6. (a) "True inflation begins only after the full employment". Explain (15)

   (b) Suppose, the price level of goods in Country A has been increased because of the increase in raw materials price. Determine what kind of inflation Country A is suffering from? Identify the policies Country A's government and central bank can take to control this kind of inflation. (20)

7. (a) Build the steps those need to be followed to calculate national income using income method and expenditure method. (15)

   (b) Define linear programming. Discuss the advantages and disadvantages of linear programming model. (15)

   (c) From the below table, estimate Gross National Product and Net National Product. (05)

Depreciation	10,000 taka
Export	1,00,000 taka
Government purchase	10,00,000 taka
Import	1,500,000 taka
Consumption	2,20,000 taka
Business investment	1,33,000 taka
Income receipts from rest of the world	12,50,000 taka
Income payment to the rest of the world	9,22,000 taka

8. (a) Distinguish between central bank and commercial bank. (10)

   (b) Suppose, Country K is suffering from deflation for the longest time. What kind of policies central bank of that country can take to stabilize the economy. Justify your opinion with suitable explanation. (15)

   (c) "Higher savings leads to a higher standard of living". Explain (10)

Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 1st Year 2nd Term Regular Examination, 2022

URP 1251

(Introduction to Architectural History)

Full Marks: 210 Time: 3.00 hrs

N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1 (a) Why was Silty clay the natural, obvious and logical choice as a building material in Bengal? (05)

(b) How do wind, water, and clay influence Bengal's settlement pattern? Explain with appropriate examples. (12)

(c) "Globalization poses a threat of homogenization and the loss of the uniqueness of architecture" – Explain the statement. (18)

2 (a) Briefly state the development of Buddhist Architecture in Bengal. (15)

(b) Write a short note on Buddhist stupa and its elements with necessary sketches. (06)

(c) Describe the architectural features of Kantaji temple with appropriate illustrations. (14)

3 (a) Briefly explain the architectural characteristics of Khan-E-Jahan style and how did the style influence the Bengal Architecture? (15)

(b) "Traditional Bengal feudal palaces has specific zoning, space organization and design criteria influenced by occidental and oriental Architecture"- describe this statement with diagrams. (20)

4 Write short notes on the following (any five-5*7) (35)

(a) Classification of temple

(b) Traditional Rekha and Pirha style

(c) Philosophy of Hindu temple

(d) Features of Mughal Architecture

(e) Basic features of Traditional Bungalow

(f) Territory of Khalifatabad

(g) Types of Buddhist Stupa

Section – B

5 (a) What is Architecture? Describe the co-relation between art and Architecture. (15)

(b) How can architecture be influenced? Discuss the influences of society, human needs and culture on architecture. (20)

6 (a) Write the geological characteristics of Egyptian architecture. (05)

(b) Delineate the evolution of the "true pyramid" from "Mastaba". Illustrate necessary sketches with it. (10)

(c) Draw the section of Djoser's Step Pyramid. (20)

7 (a) What is Order? Discuss the 3 types of Greek order with proper sketches. (20)

(b) What is Aqueduct? Illustrate the difference between Roman and Modern aqueduct. (15)

8 (a) Describe the ideologies of Modern Architecture. (10)

(b) Although the emergence of Modern Architecture happened in a decisive historical moment, the movement failed discouragingly. Write the reasoning of this failed phenomenon. (20)

(c) Narrate the notion – “Form follows Function”. (05)

Khulna University of Engineering & Technology

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BURP 1st Year 2nd Term Regular Examination, 2022

URP-1281

(Surveying and Cartography)

Full Marks: 210 Time: 3.00 hrs

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define the following with necessary sketches and examples: (20)
   i) Selection of survey stations
   ii) Well-conditioned and ill-conditioned triangles
   iii) Reconnaissance survey and index sketch
   iv) Single-line and double-line field book

   (b) What are the methods of plane tabling? Describe them with necessary sketch. (15)

2. (a) Define the following with necessary sketch and example: (20)
   i) True meridian and magnetic meridian
   ii) Whole circle bearing and reduced bearing.
   iii) Isogonic and agonic line
   iv) Variation of magnetic declination

   (b) A closed traverse is conducted with five stations. A, B, C, D and E taken in anticlockwise order, in form of a regular pentagon. If the FB of AB is $37^0 30'$, find the FBs of the other sides. (15)

3. (a) The following consecutive readings were taken with a level and a 4-meter leveling staff on a continuously sloping ground at common intervals of 25 m: (20)
   2.775 m (on A), 1.501 m, 1.322 m, 2.127 m, 2.811 m, 3.055 m, 3.366 m, 2.055 m, 2.433 m, 0.519 m, 1.589 m, 2.700 m, 3.500 m and 3.363 m (on B).
   The RL of A was 445.500 m. Make entries in a level book and apply the usual checks. The change points were taken at 75 m and 200 m in chainage.

   (b) What types of errors may occur for the short segments of leveling? - Explain briefly. (15)

4. (a) What is contour line? Explain the characteristics of contours with necessary sketches. (15)

   (b) Define the terms “contour interval” and “horizontal equivalent”. (10)

   (c) The following perpendicular offsets were taken from chain line to an irregular boundary at 15 m interval: 7.55, 2.80, 3.35, 5.46, 2.22, 7.80, 8.95, 2.52, 6.13 and 5.20 m. Calculate the area between chain line, the boundary and the end offsets by trapezoidal rule. (10)

Section – B

5. (a) Maps allow many types of relationships to be formed. Give examples of the following: (15)
   - Relationships among locations with no attributes
   - Relationship among various attributes at the same point
   - Relationship among different locations of the same attribute
   - Relationships among locations of combined attributes of given distributions
   - Relationships among attributes with no location.

   (b) “A map is not a photograph” - explain this statement with proper example. (10)

   (c) Why should maps be made easy to interpret? Give an example of a notable map that adheres to this philosophy. (10)

6. (a) Describe the following with proper illustrations: (20)
   I. Choropleth map
   II. Proximal map
   III. Isarithmic
   IV. Symbol map

   (b) List the elements of a map composition and provide examples of a bad and good map layout. (15)

7. (a) Compare the following: (20)
   I. Datum vs Geoid vs Ellipsoid
   II. Coordinate Systems vs Latitude and Longitude
   III. Coordinate Systems vs. Map Projections
   IV. Cylindrical vs Conic vs Planar Projections

   (b) Describe the various types of distortions in Map projections. (10)

   (c) Why is BTM preferable than UTM in Bangladesh? (05)

8. (a) Explain Georeferencing. (15)

   (b) What do you understand by map projections? Explain the different types of distortion with relation to the basic map projection. (15)

   (c) For the Earth, a = 1750 km; b = 2555 km. Calculate the flattening ratio. (05)

Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 1st Year 2nd Term Regular Examination, 2022

URP 1211

(Urban Planning Principles)

Full Marks: 210 Time: 3.00 hrs

N.B. i) Answer any three questions from each section in separate script.

ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Khulna Development Authority (KDA) is responsible for town planning in Khulna city. Discuss in brief the scope of their work. (15)

   (b) A real estate company is planning to build a three-story building on a rectangular plot that has 200 feet of street frontage and 300 feet of depth. The first, second and third story measures 100 feet by 200 feet. Calculate the FAR of the building. (10)

   (c) "A well-planned bus terminal acts as catalyst to social and economic development of surrounding areas"- how so? Explain. (10)

2. (a) "Transit Oriented Development is a very much practical way of neighborhood planning rather than Radburn Concept"- do you agree? State your opinion with necessary diagram. (15)

   (b) Why the function of a well-planned town is similar to the function of a living organism? Discuss in brief in essence with the objectives of town planning. (10)

   (c) Write down the major differences between satellite town and new town. (10)

3. (a) Briefly discuss the characteristics of a planned neighborhood unit. (10)

   (b) Briefly discuss the suburban neighborhood unit concept and specify how suburban neighborhood unit can increase environmental pollution. (15)

   (c) "New town can promote regional balance"- discuss in brief. (10)

4. (a) Write down the evolvement process of satellite town to new town. (15)

   (b) "Airport creates a positive influence on Land Development Patterns" – explain. (05)

   (c) Write a short note on any three of the following: (15)

      i) Boulevard or Parkways

      ii) National Park

      iii) Greenbelt

      iv) Zoning

Section – B

5. (a) What is your understanding about "Primate City"? Explain the conditions responsible for the growth of primate city with necessary examples. (10)

   (b) What do you mean by Urban System? Describe the law of rank size rule with necessary example. (10)

   (c) Briefly explain the three principles of "Central Place Theory" with necessary diagram. (15)

6. (a) Discuss various types of towns based on function with necessary examples. (15)

   (b) What do you mean by town center? Illustrate any five principles required for developing a successful town center. (15)

   (c) Briefly explain your understanding of the term "Urban Periphery". (05)

7. (a) Briefly discuss the classifications of industry according to nature of processing and types of products with necessary examples. (12)

   (b) "Vehicular mobility gradually increases and land access decreases from local roads to arterials"- Explain using necessary diagram. (08)

   (c) Provide a brief discussion on the functions of the median, shoulder and lay-by of a roadway. (15)

8. (a) Write a short note on the following: (10)

   (i) Urban Pyramid

   (ii) Normal formula method

   (b) Present a case study illustrating the concept of industrial agglomeration and subsequently provide a concise analysis of both its benefits and drawbacks. (15)

   (c) "Eco-industrial parks seek to minimize energy use, waste generation, operating cost and other environmental impacts" – Justify the statement. (10)
